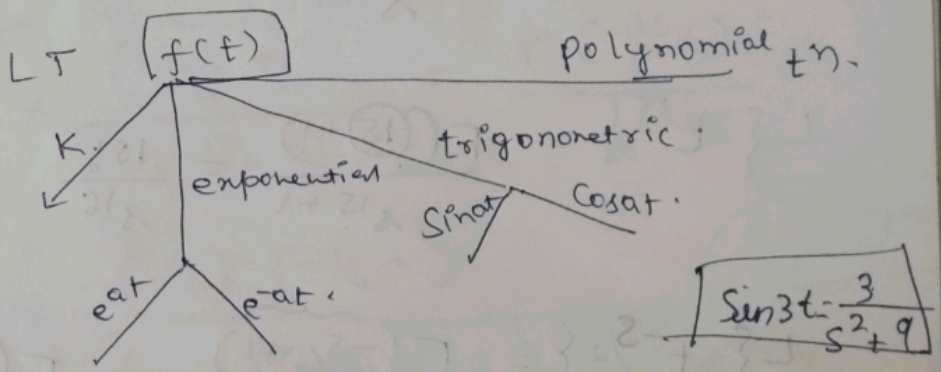
To solve differential equation.

↓ rule → Differential equation & Intigral are converted to Algebraic, (converting to simple form)

$f(t)$ $t \geq 0$, s

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt \rightarrow \text{finite exists} = F(s)$$

① constant, e^{at}, e^{-at}.



$$* \; L(k) = \frac{k \cdot 1}{s^1} = \frac{1}{s}, \quad * \; L(e^{at}) = \frac{1}{s-a}$$

$$* \; L(e^{-at}) = \frac{1}{s+a} \quad * \; L(\sin at) = \frac{a}{s^2+a^2}$$

$$* \; \cos(at) = \frac{s}{s^2+a^2} \quad * \; L\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}} \quad \frac{n\Gamma(n)}{s^{n+1}} \quad \frac{n!}{s^{n+1}}$$

$$L\{t^6\} = \frac{\Gamma(6+1)}{s^{6+1}} = \frac{6!}{s^7}$$

$$L\{t^{5/2}\} = \frac{\Gamma(5/2+1)}{s^{5/2+1}} = \frac{5}{2}\frac{\Gamma(5/2)}{s^{5/2+1}} \quad \frac{3}{2}\Gamma\frac{3}{2} \quad \Gamma(3/2+1)$$

$$\mathcal{L}\{t^{5/2}\} = \frac{\Gamma(5/2+1)}{s^{5/2+1}} = \frac{\frac{5}{2}\Gamma(5/2)}{s^{3/2+1}} = \frac{\frac{5}{2}\cdot\frac{3}{2}\Gamma(3/2)}{s^{1/2+1}}$$

$$= \frac{\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2)}{s^{7/2}}$$

$$\mathcal{L}\{t^{11/2}\} = \frac{\Gamma(\frac{11}{2}+1)}{s^{11/2+1}} = \frac{\frac{11}{2}\cdot\frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2)}{s^{13/2}}$$

$$\mathcal{L}\{t^{13/4}\} = \frac{\Gamma(\frac{13}{4}+1)}{s^{13/4+1}} = \frac{\frac{13}{4}\cdot\frac{9}{4}\cdot\frac{5}{4}\cdot\frac{1}{4}\Gamma(1/4)}{s^{17/4}}$$

$$\mathcal{L}\{t^{15}\} = \frac{\Gamma(15+1)}{s^{15+1}} = \frac{15!}{s^{16}}$$

$$\mathcal{L}\{t^{-5/2}\} = \frac{\Gamma(-5/2+1)}{s^{-5/2+1}} = \frac{\Gamma(-3/2)}{s^{-3/2}} = \frac{\Gamma(-3/2+1)}{-\frac{3}{2}\,s^{-3/2}}$$

$\Gamma(n+1) = n!$ — n +ve integer $= \dfrac{\Gamma(-1/2)}{-\frac{3}{2}\,s^{-3/2}}$

$n\Gamma(n)$ — n +ve fraction

$\Gamma(n)$ — n is −ve fraction $= \dfrac{\Gamma(-1/2+1)}{(-\frac{1}{2})(-\frac{3}{2})\,s^{-3/2}}$

$\Gamma(n) = \dfrac{\Gamma(n+1)}{n}$

$$\mathcal{L}\{t^{-5/2}\} = \frac{\Gamma(-5/2+1)}{s^{-5/2+1}} = \frac{\Gamma(-3/2)}{s^{-3/2}}$$

$$= \frac{\Gamma(-1/2+1)}{\left(\frac{-3}{2}\right)\left(\frac{-1}{2}\right) s^{-3/2}}$$

$$= \frac{\sqrt{\pi}}{\frac{3}{4}\, s^{-3/2}}$$

$$\mathcal{L}\{t^{-17/4}\} = \frac{\Gamma(-\frac{17}{4}+1)}{s^{-17/4+1}} = \frac{\Gamma(-13/4)}{s^{-13/4}}$$

$$= \frac{\Gamma(-1/4+1)}{\left(\frac{-13}{4}\right)\left(\frac{-9}{4}\right)\left(\frac{-5}{4}\right)\left(\frac{-1}{4}\right) s^{-13/4}}$$

Till proper fraction

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① $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

$\mathcal{L}\{e^{-at}\} = \frac{1}{s+a}$

exponential forms

Sinhat $= \frac{e^{at}-e^{-at}}{2}$

Coshat $= \frac{e^{at}+e^{-at}}{2}$

$e^{\log a^t} = t\log a$

$= \frac{1}{s-\log}$

① $e^{2t} + 3e^{-t} + 5$

$= L\{e^{2t}\} + L\{3e^{-t}\} + L\{5\}$

$= \frac{1}{s-2} + \frac{3}{s+1} + \frac{5}{s}$

② $\sinh^2 3t + \cosh^3 2t + (e^t + 5)^2$

$= \left[\frac{e^{3t} - e^{-3t}}{2}\right]^2 + \left[\frac{e^{2t} + e^{-2t}}{2}\right]^3 + (e^t)^2 + 5^2 + 10e^t$

$= \frac{1}{4}\left[(e^{3t})^2 + (e^{-3t})^2 - 2e^{3t}\cdot e^{-3t}\right] + \frac{1}{8}\left[(e^{2t})^3 + (e^{-2t})^3 + 3e^{2t}e^{-2t}(e^{2t} + e^{-2t})\right]$

$+ e^{2t} + 25 + 10e^t$

$\Rightarrow \frac{1}{4}\left[e^{6t} + e^{-6t} - 2\right] + \frac{1}{8}\left[e^{6t} + e^{-6t} + 3e^{2t} + 3e^{-2t}\right]$

$+ e^{2t} + 25 + 10e^t$

$= \frac{1}{4}\left[\frac{1}{s-6} + \frac{1}{s+6} - \frac{2}{s}\right] + \frac{1}{8}\left[\frac{1}{s-6} + \frac{1}{s+6} + \frac{3}{s-2} + \frac{3}{s+2}\right] + \frac{1}{s-2} + \frac{25}{s}$

$$\mathcal{L}\left\{e^{\log 2^{3t}} + 5 + \log 2\right\}$$

$$e^{\log 2^{3t}} = e^{3t \log 2}$$

$$= \frac{1}{s - 3\log 2} + \frac{5}{s} + \frac{\log 2}{s}$$

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2} \qquad \mathcal{L}\{\sinh at\} = \frac{a}{s^2 - a^2}$$

$$\mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2} \qquad \mathcal{L}\{\cosh at\} = \frac{s}{s^2 - a^2}$$

Trigonometric forms

(1) $\sin^2\theta = \frac{1 - \cos 2\theta}{2}$

(2) $\cos^2\theta = \frac{1 + \cos 2\theta}{2}$

(3) $\sin^3\theta = \frac{3\sin\theta - \sin 3\theta}{4}$

(4) $\cos^3\theta = \frac{3\cos\theta + \cos 3\theta}{4}$

(5) $\sin A \cos B = \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]$

(6) $\sin(-\theta) = -\sin\theta$

(7) $\cos(-\theta) = \cos\theta$

(8) $\sin A \sin B = -\frac{1}{2}\left[\cos(A+B) - \cos(A-B)\right]$

(9) $\cos A \cos B = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]$

① $\sin^2(5t)$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\sin^2(5t) = \frac{1-\cos 10t}{2}$

$= \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+10^2}\right]$

$\cos^2 3t$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$\cos^2(3t) = \frac{1+\cos 6t}{2}$

$= \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+6^2}\right]$

$\sin 3t \cos 2t$.

$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

$\sin 3t \cos 2t = \frac{1}{2}[\sin(5t) + \sin(t)]$

$= \frac{1}{2}\left[\frac{5}{s^2+5^2} + \frac{1}{s^2+1}\right]$

$\cos 3t \sin 2t$.

$\sin 2t \cos 3t = \frac{1}{2}[\sin 5t + \sin(-t)]$

$= \frac{1}{2}[\sin 5t - \sin t]$

—

$\cos(2t+3)$

$\Rightarrow \cos(A+B) = \cos A \cos B - \sin A \sin B$

$= \cos 2t \cos 3 - \sin 2t \cdot \sin 3$

$= (\cos 3)\left(\frac{s}{s^2+2^2}\right) - (\sin 3)\left(\frac{2}{s^2+2^2}\right)$

$\sin t \sin 5t$

$\sin A \sin B = -\frac{1}{2}\left[\cos(A+B) - \cos(A-B)\right]$

$= -\frac{1}{2}\left[\cos 6t - \cos(-4t)\right]$

$= -\frac{1}{2}\left[\cos 6t - \cos 4t\right]$

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(1) $\sin^2 5t$ (2) $\cos^2 3t$.

(3) $\sin 9t \cos 7t$ (4) $\sin 4t \sin 7t$

(5) $\cos 3t \cos 5t$. (6) $\sin^3 8t$.

(7) $\cos^3 2t$.

① $\sin^2 5t$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\sin^2 5t = \frac{1-\cos 5t}{2}$

$\mathcal{L}\{\} = \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+25}\right]$ //

② $\cos^2 3t$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$= \frac{1+\cos 6t}{2}$

$\mathcal{L}\{\} = \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+36}\right]$

did not Multiple 2 to θ

③ $\sin 9t \cos 7t$

$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

$9t \quad 7t = \frac{1}{2}[\sin 16t + \sin 2t]$

$= \frac{1}{2}\left[\frac{16}{s^2+16^2} + \frac{2}{s^2+4}\right]$

④ $\sin 4t \sin 7t$

$\sin A \sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$

$= \frac{1}{2}[\cos(11t) - \cos(-3t)]$

$= \frac{1}{2}[\cos(11t) - \cos(3t)]$

$= \frac{1}{2}\left[\frac{s}{s^2+11^2} - \frac{s}{s^2+3^2}\right]$

⑤ $\cos 3t \cos 5t = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$

$= \frac{1}{2}[\cos(8t) + \cos(2t)]$

$= \frac{1}{2}\left[\frac{s}{s^2+8^2} + \frac{s}{s^2+2^2}\right]$

cos → formula

⑥ $\sin^3 8t = \dfrac{3\sin\theta - \sin 3\theta}{4}$

$= \dfrac{3\sin 8t - \sin 24t}{4}$

$= \dfrac{1}{4}\left[\dfrac{3\times 8}{s^2+8^2} - \dfrac{24}{s^2+24^2}\right]$

$= \dfrac{1}{4}\left[\dfrac{24}{s^2+8^2} - \dfrac{24}{s^2+24^2}\right]$

⑦ $\cos^3 2t = \dfrac{3\cos\theta + \cos 3\theta}{4}$

$= \dfrac{3\cos 2t + \cos 6t}{4}$

$= \dfrac{1}{4}\left[\dfrac{3s}{s^2+4^2} + \dfrac{s}{s^2+6^2}\right]$

$L\{f(t)\} = F(s)$

Properties

division: $t \cdot L\left\{\dfrac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$

Scaling

Shifting

Multiplication: $t^n L\{t^n f(t)\} = (-1)^n \dfrac{d^n F(s)}{ds^n}$

$L\{f(kt)\} = \dfrac{1}{k} F\left(\dfrac{s}{k}\right)$

$L\{f(t/k)\} = k\,F(ks)$

$L\left\{f\left(\dfrac{2t}{3}\right)\right\} = \dfrac{3}{2} F\left(\dfrac{3s}{2}\right)$

$e^{at} f(t) = F(s-a)$

$e^{-at} f(t) = F(s+a)$

$n=1$: $L\{t f(t)\} = -\dfrac{d}{ds} F(s)$

$n=2$: $L\{t^2 f(t)\} = +\dfrac{d^2}{ds^2} F(s)$

If $L\{f(t)\} = \frac{10s+3}{s^2+9} = F(s)$

then find

① $L\{f(3t)\} = \frac{1}{3}\left[\frac{10\left(\frac{s}{3}\right)+3}{\left(\frac{s}{3}\right)^2+9}\right]$

② $L\{f(t/3)\} = 3\left[\frac{10(3s)+3}{(3s)^2+9}\right]$

③ $L\{f(\frac{2t}{3})\} = \frac{3}{2}\left[\frac{10\left(\frac{3}{2}s\right)+3}{\left(\frac{3s}{2}\right)^2+9}\right]$

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① $e^{4t}\sin^2 3t$

for $f(t) = \sin^2(3t)$

$f(t) = \frac{1-\cos 6t}{2}$ $\left[\sin^2\theta = \frac{1-\cos 2\theta}{2}\right]$

$L\{f(t)\} = \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+36}\right]$

$\therefore L\{e^{4t}\sin^2 3t\} = \frac{1}{s-4} - \frac{s-4}{(s-4)^2+36}$

$s \to s-4$

② $e^{-7t}\cos^3 5t$

Let $f(t) = \cos^3(5t)$

$f(t) = \frac{3\cos 5t + \cos 15t}{4}$ $\left[\cos^3\theta = \frac{3\cos\theta + \cos 3\theta}{4}\right]$

$L\{f(t)\} = \frac{1}{4}\left[\frac{3s}{\quad} + \frac{s}{\quad}\right]$

$s \to s+7$

$\therefore L\{e^{-7t}\cos^3 5t\} = \frac{1}{4}\left[\frac{3(s+7)}{(s+7)^2+25} + \frac{(s+7)}{(s+7)^2+5^2}\right]$

Shifting property of Exponential.

t

$L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$

$L\{t^2 \cos 3t\}$

Let $f(t) = \cos 3t$.

$L\{f(t)\} = \frac{s}{s^2+9} = F(s)$

$\therefore L\{t^2\cos 3t\} = (-1)^2 \frac{d^2}{ds^2} F(s)$

$= + \frac{d}{ds}\frac{d}{ds}\left[\frac{s}{s^2+9}\right]$

$= \frac{d}{ds}\left[\frac{(s^2+9)\cdot 1 - s\cdot(2s)}{(s^2+9)^2}\right]$

$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

$= \frac{d}{ds}\left[\frac{s^2+9-2s^2}{(s^2+9)^2}\right]$

$= \frac{d}{ds}\left[\frac{-s^2+9}{(s^2+9)^2}\right]$

$= \left[\frac{(s^2+9)^2(-2s) - (-s^2+9)\,2(s^2+9)2s}{[(s^2+9)^2]^2}\right]$

$2s(s^2+9)\left[(s^2+9)(-1) - (-s^2+9)2\right]$

$(s^2+9)^4$

$2s\left[-s^2-9+2s^2-18\right]$

$(s^2+9)^3$

$t^2 \sin 4t$

$f(t) = \sin 4t$

$L\{f(t)\} = \dfrac{4}{s^2+16} = F(s)$

$\therefore L\{t^2 \sin 4t\} = (-1)^2 \dfrac{d^2F(s)}{ds^2}$

$= + \dfrac{d}{ds}\dfrac{d}{ds}\left[\dfrac{4}{s^2+16}\right]$

$= 4 \dfrac{d}{ds}\left[\dfrac{-1}{(s^2+16)^2} \cdot 2s\right]$ (1)

$= -8 \dfrac{d}{ds}\left[\dfrac{s}{(s^2+16)^2}\right]$

$= -8 \left[\dfrac{(s^2+16)^2 \cdot 1 - s \cdot 2(s^2+16) 2s}{[(s^2+16)^2]^2}\right]$ (2)

$= -8(s^2+16)\left[\dfrac{s^2+16-4s^2}{(s^2+16)^3}\right]$

$= -8\left[\dfrac{16-3s^2}{(s^2+16)^3}\right]$

$\dfrac{d}{dx}\left[\dfrac{1}{x^2}\right] = \dfrac{-1}{x^2}$

$\dfrac{d}{dx}\left[\dfrac{1}{x^8}\right] = \dfrac{-8}{x^9}$

Q) $t \sin 5t$ Q) $t \cos 8t$

~~Let $\sin 5t = F(s)$~~

consider $\sin 5t = s$

$$\mathcal{L}(\sin 5t) = \frac{5}{s^2+25}$$

so $\mathcal{L}\{t \sin 5t\} = \frac{d}{ds}\left(\frac{5}{s^2+25}\right) = \frac{(s^2+25).0 - 5(2s)}{(s^2+25)^2}$

$$= \frac{(5^2+s^2)\times 0 - 5\times 2(s^2+5^2)2s}{(s^2+5^2)^2}$$

$$= \frac{-20s(s^2+5^2)}{(s^2+5^2)^2}$$

$$= \frac{-20s}{(s^2+5^2)}$$

Denominator should be taken from qution,
The previous expression had $(Dr)^2$

Q) $t \cos 8t$

consider $\cos 8t$

$$\mathcal{L}(\cos 8t) = \frac{s}{s^2+8^2}$$

$(-1)^n \frac{d^n}{ds^n} F(s)$

$n \rightarrow$ Number of time diff is carried out

$$\therefore \mathcal{L}(t \cos 8t) = \frac{d}{ds}\left(\frac{s}{s^2+8^2}\right)$$

$$= \frac{(s^2+8^2).1 - s(2(s^2+8^2)2s}{(s^2+8^2)^2}$$

$$= \frac{(s^2+8^2) - 2s^2}{(s^2+8^2)^2}$$

$$= \frac{(s^2+8^2)[1] - 8s^2}{(s^2+8^2)^2}$$

$$= \frac{8^2 - s^2}{(s^2+8^2)^2}$$

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$$

Q $\dfrac{e^{3t} - e^{-6t}}{t}$

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* $\int \frac{1}{s \pm a}\,ds = \log(s-a)$

$s^2 \pm a^2 = t$

$2s\,ds = dt$

$= \frac{1}{2}\,dt$

* $\int \frac{s}{s^2 \pm a^2}\,ds = \frac{1}{2}\log(s^2+a^2)$

* $\int \frac{1}{s^2+a^2}\,ds$ ~~$\frac{1}{a}\tan^{-1}\left(\frac{s}{a}\right)$~~ $\Rightarrow \int \frac{1}{t}\,\frac{dt}{2}$

* ~~$\int \frac{1}{s}\,ds = \log s$~~

$$L\left\{\frac{e^{3t} - e^{-6t}}{t}\right\}$$

Let $f(t) = e^{3t} - e^{-6t}$

$$L\{f(t)\} = \frac{1}{s-3} - \frac{1}{s+6} = F(s)$$

$$\therefore L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds = \int_s^\infty \left[\frac{1}{s-3} - \frac{1}{s+6}\right] ds$$

$$= \Big[\log(s-3) - \log(s+6)\Big]_s^\infty$$

$$= \left[\log\left[\frac{s-3}{s+6}\right]\right]_s^\infty$$

$$= \log\left[\frac{s(1-3/s)}{s(1+6/s)}\right]_s^\infty$$

$$= \log 1 - \log\left(\frac{1-3/s}{1+6/s}\right)$$

$$= 0 - \log\left[\frac{\frac{s-3}{s}}{\frac{s+6}{s}}\right]$$

$$= -\log\left(\frac{s-3}{s+6}\right)$$

$$= \log\left(\frac{s+6}{s-3}\right)$$

$$L\left\{\frac{1-e^{-at}}{t}\right\}$$

Let $f(t) = 1 - e^{-at}$

$$L\{f(t)\} = \frac{1}{s} - \frac{1}{s+a} = F(s)$$

$$\therefore L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds = \int_s^\infty \left[\frac{1}{s} - \frac{1}{s+a}\right] ds$$

$$= \left[\log s - \log(s+a)\right]_s^\infty$$

$$= \left[\log\left[\frac{s}{s+a}\right]\right]_s^\infty$$

$$= 0 - \log\left(\frac{s}{s+a}\right)$$

$$= \log\left(\frac{s+a}{s}\right)$$

$$L\left\{\frac{\sin 3t}{t}\right\}$$

Let $f(t) = \sin 3t$

$$L\{f(t)\} = \frac{3}{s^2+9} = F(s)$$

$$\therefore L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds = \int_s^\infty \frac{3}{s^2+9^2}\,ds$$

$$= \left[\frac{1}{3}\tan^{-1}\left(\frac{s}{3}\right)\right]_s^\infty$$

$$= \tan^{-1}\infty - \tan^{-1}(s/3)$$

$$= \pi/2 - \tan^{-1}(s/3)$$

$$= \cot^{-1}(s/3) \qquad \left[\tan^{-1}x + \cot^{-1}x = \pi/2\right]$$

$$\frac{\sin 3t \sin 5t}{t}$$

Let $f(t) = \sin 3t \sin 5t$

$$f(t) = -\frac{1}{2}\left[\cos 8t - \cos 2t\right] \qquad \left[\sin A \sin B = -\frac{1}{2}\left[\cos(A+B) - \cos(A-B)\right]\right]$$

$$L\{f(t)\} = -\frac{1}{2}\left[\frac{s}{s^2+8^2} - \frac{s}{s^2+2^2}\right] = F(s)$$

$$\therefore L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds = -\frac{1}{2}\int_s^\infty \left(\frac{s}{s^2+8^2} - \frac{s}{s^2+4}\right)ds$$

$$= -\frac{1}{2}\left[\frac{1}{2}\log(s^2+8^2) - \frac{1}{2}\log(s^2+4)\right]$$

$$= -\frac{1}{4}\left[\log(s^2+8^2) - \log(s^2+4) \right]_s^\infty$$

$$= -\frac{1}{4}\left[\log\left(\frac{s^2+8^2}{s^2+4}\right) \right]_s^\infty$$

$$= -\frac{1}{4}\left[0 - \log\left(\frac{s^2+8^2}{s^2+4}\right) \right]$$

$$= \frac{1}{4}\log\left(\frac{s^2+8^2}{s^2+4}\right)$$